Rudra Patel

Edmonton, Alberta
rudrapatel1440@gmail.com

EDUCATION

University of Alberta — Edmonton,AB *Sept 2025 - Expected June 2029*

Bachelor of Computer Science, Honors

- Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Linear Algebra, Probability, Statistics, Calculus

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, Assembly, SQL, Bash
Machine Learning: NumPy, Pandas, OpenCV
Software Engineering: Git, GitHub, Linux, MongoDB
Robotics / Systems: ROS 2, C++, Raspberry Pi, Linux

EXPERIENCE

Mathnasium — Math Instructor *Sept 2026 – Present*

- Taught Math 10, 20, and 30 to individuals and small groups, following the Alberta Curriculum guidelines.

HSC Technologies — IT Support *May 2026 – Aug 2026*

- Provided remote technical support to 100+ global employees, diagnosing and resolving VPN, Wi-Fi, and network connectivity issues via chat, phone, and ticketing systems.
- Utilized remote desktop software to securely access user devices, troubleshoot configuration errors, and optimize network settings.
- Maintained documentation for common remote-work network fixes, reducing average ticket resolution time by 15%.

Indian Rocketry Association (IRA) - Team Member *May 2025 – Aug 2025*

- Rocket Control Systems & Embedded Software — Collaborated with a multidisciplinary team to develop and test embedded software for rocket flight-control systems, working with sensors, microcontrollers, and real-time control logic.

PROJECTS

Neural Network From Scratch

Python Numpy

- Built a fully connected neural network from scratch in NumPy (forward/backpropagation, sigmoid activations, mini-batch gradient descent), reaching 97% test accuracy on MNIST.
- Implemented and compared squared-error and cross-entropy losses; tuned learning rate and hidden layer size to improve accuracy from 52% to 97%.

Chess AI

Python, Minimax, Alpha-Beta Pruning

- Built a chess-playing AI using the Minimax algorithm to evaluate possible game states and select moves.

HTTP Server From Scratch

C++, Linux, Sockets, Multithreading

- Built a multithreaded HTTP server in C++ on Linux using POSIX sockets, handling multiple concurrent client connections.

CERTIFICATIONS

- CS50's Introduction to Artificial Intelligence with Python — Harvard University
- Supervised Machine Learning: Regression and Classification — Stanford Online
- CCNA Networking – In Progress